$\Rightarrow \boxed{b-\varepsilon < z_n < b+\varepsilon}$ (Eqn 4)

- Take $N = \max(N_1, N_2)$. $\forall n > N$, Eqn(2) and Eqn(4) hold.

$\boxed{a+b-2\varepsilon < y_n + z_n < a+b+2\varepsilon} \longrightarrow$ Eqn(2) + Eqn(4) = Eqn(5)

or, $(a+b)-\varepsilon_1 < y_n+z_n < (a+b)+\varepsilon_1 \Rightarrow \boxed{|(y_n+z_n)-(a+b)| < \varepsilon_1}$. (Eqn 6)

where $\varepsilon_1 = 2\varepsilon$

This Eqn (6) holds for any arbitrary ε. $\boxed{\therefore (y_n+z_n) \text{ converges to } (a+b)}$ □

Reader: If we select an arbitrary $\varepsilon > 0$, then there is a number N_1 such that for all the terms of the first sequence with $n > N_1$ we shall have $|y_n - a| < \varepsilon$. In addition, for the same ε there is N_2 such that for all the terms of the second sequence with $n > N_2$ we shall have $|z_n - b| < \varepsilon$. If now we select the greatest among N_1 and N_2 (we denote it by N), then for all $n > N$ both $|y_n - a| < \varepsilon$ and $|z_n - b| < \varepsilon$. Well, this is as far as I can go.

Author: Thus, you have established that for an arbitrary ε there is N such that for all $n > N$ both $|y_n - a| < \varepsilon$ and $|z_n - b| < \varepsilon$ hold simultaneously. And what can you say about the modulus $|(y_n+z_n)-(a+b)|$ (for all n)?

I remind you that $|A+B| \leq |A| + |B|$.

Reader: Let us look at $|(y_n+z_n)-(a+b)| = |(y_n-a)+(z_n-b)| \leq [|y_n-a| + |z_n-b|] < (\varepsilon+\varepsilon) = 2\varepsilon$

Author: You have proved the theorem, haven't you?

Reader: But we have only established there is N such that for all $n > N$ we have $|(y_n+z_n)-(a+b)| < 2\varepsilon$. But we need to prove that $|(y_n+z_n)-(a+b)| < \varepsilon$.

Author: Ah, that's peanuts, if you forgive the expression. In the case of the sequence (y_n+z_n) you select a value of ε, but for the sequences (y_n) and (z_n) you must select a value of $\frac{\varepsilon}{2}$ and namely for this value find N_1 and N_2.

Thus, we have proved that if the sequences (y_n) and (z_n) are convergent, the sequence (y_n+z_n) is convergent too. We have even found a limit of the sum. And do you think that the converse is equally valid?